AI FOR SUSTAINABLE & RESILIENT SUPPLY CHAINS

PAPER 1
INVARIANT-SC

Auditable Semantic Recovery for Sustainable and Resilient Supply Chains

CONCEPT IN ONE SENTENCE — Core Postulate

Recover what supply-chain records mean before asking AI to optimize them: learn a common, auditable interpretation of heterogeneous enterprise data using **physical flow consistency**, while preserving evidence and explicitly representing meaning that the observations cannot resolve.

Physical Invariants
Mass, energy & temporal conservation laws provide self-supervisory ground truth.

Zero-Shot Recovery
Transfers to unseen schemas and ERPs without manual table-by-table mapping.

Lineage & Ambiguity
Explicit non-uniqueness: abstains or surfaces alternatives when evidence is sparse.

4 Downstream Tasks
One recovered flow foundation powering Carbon, Circularity, Inventory & Resilience.

RESEARCH IDENTITY
One new learning problem, one canonical scientific object, multiple coordinated downstream tests.

DISCLOSURE LEVEL
Problem statement, contribution boundaries, evaluation logic, and real-world value; sensitive mechanics withheld.

01 The Real Problem

DATA HETEROGENEITY & STRUCTURAL BLINDSPOTS

Most supply-chain AI begins after records have already been cleaned, joined, and given stable meanings. **That assumption is routinely false across real-world organizations.**

Procurement, warehouse, manufacturing, transport, returns, and finance systems frequently encode the identical physical event with incompatible schemas, conflicting units, opposite signs, disjoint identifiers, mismatched granularity, and asynchronous posting times. A receipt in one warehouse database may appear as a ledger adjustment in another; a single intermodal shipment is routinely fractured across several decoupled tables.

This is not merely an IT integration inconvenience. If activity records are interpreted incorrectly, every machine learning model, optimization routine, and reporting engine built above them inherits the corruption: **Scope 3 activity data becomes impossible to audit**, material losses and circular returns are misclassified, inventory states drift silently, and disruption paths are reconstructed from an unstable foundation. The National Institute of Standards and Technology (NIST) similarly identifies cross-domain semantic differences and low-effort traceability-data integration as a critical unresolved national need [5].

⚠ **THE HIDDEN PREREQUISITE**

Before a system can forecast, optimize, calculate emissions, or plan disaster recovery, it must first know **what each record means** and verify whether all enterprise records can **jointly describe a physically plausible process**.

1.1 Why Existing Fixes Are Incomplete

1 Manual Mappings & Rules
Prohibitively expensive, brittle under schema changes, and impossible to audit as enterprise software evolves.

2 LLM & Semantic Matching
Can match plausible column labels based on text tokens, but cannot guarantee a globally coherent physical flow [1].

3 Traceability Standards (EPCIS)
Define rigid destination schemas [5, 6], yet arbitrary legacy records must still be reconciled into them.

4 Relational Deep Learning
Assumes tables, primary/foreign keys, and relational edges are already meaningful and ground-truth [2].

5 Commercial Dashboards & Aggregators
Visualize integrated data downstream; they cannot establish whether the underlying data reconciliation is scientifically defensible or physically viable.

CENTRAL RESEARCH QUESTION (PAPER 1)

Can an AI system recover a shared supply-chain interpretation from heterogeneous records by exploiting coupled physical-flow regularities, and can it identify when the evidence supports several interpretations rather than one?

02 One Umbrella, One Scientific Object

ARCHITECTURAL PARADIGM

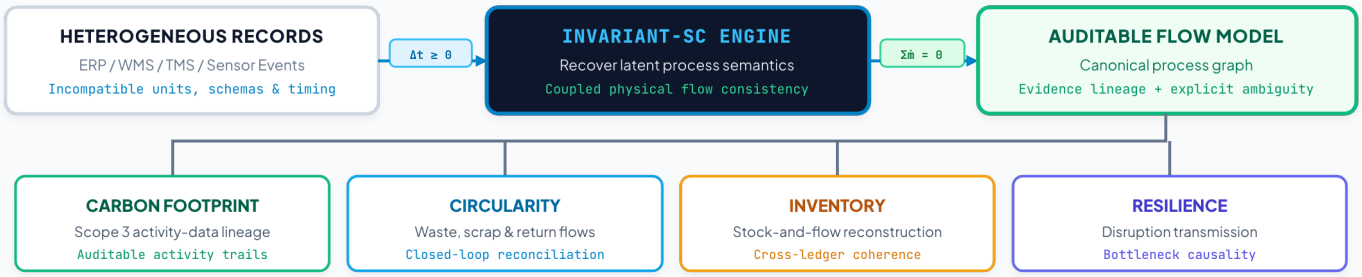


Figure 1. Every application queries the identical recovered flow model; there are zero independent feature silos or bespoke ETL pipelines.

2.1 The Core Mechanism

INVARIANT-SC treats enterprise data reconciliation as a structured learning problem. It seeks a canonical description of entities, movements, transformations, stocks, timing, and measurement conventions that makes many records jointly coherent. The essential supervision comes from **regularities of the underlying physical process**, not merely from fragile table headers or documentation.

2.2 The Restrained Evidence Layer

The recovered model carries explicit source lineage and evidence consistency. When two materially different meanings fit observations, the system preserves alternatives or **abstains instead of manufacturing false certainty**. This ambiguity-aware function strengthens the core claim without bloating the paper into a separate product.

2.3 Why This Is a Scientifically Valid Umbrella

Downstream Question	Query on the Shared Recovered Model	Methodological Boundary
Carbon Footprint	Which traceable material, energy, and transport activities occurred across tiers?	Emission factors and reporting standards remain external.
Waste / Circularity	Where did material leave, return, transform, or fail to reconcile in the network?	No claim to solve macro circular-economy policy questions.
Reconciled Inventory	What stock trajectory is mathematically supported across inconsistent ledgers?	The system explicitly exposes unsupported, drifting intervals.
Supply Resilience	Which physical flows and operational dependencies transmit an upstream disruption?	Downstream decision optimization is an external extension.

UNIFYING THESIS

Recover an auditable set of **physically consistent canonical supply-chain interpretations** from heterogeneous enterprise records, and explicitly separate what is **supported**, **ambiguous**, or **missing**.

03 Where the Novelty Actually Lives

PRIOR-ART DIFFERENTIATION & THEORETICAL MOAT

The novelty is not the generic phrase *"AI for supply chains,"* and it is not the superficial act of combining carbon and resilience metrics on one screen.

The genuine contribution is the proposed **learning object and self-supervision mechanism**: inferring latent enterprise data meaning and multi-echelon process structure from **coupled physical consistency**, then returning an auditable interpretation set rather than a forced, brittle one-to-one mapping.

3.1 Comparative Positioning vs. Adjacent Research Fields

Adjacent Direction	What It Normally Assumes or Solves	INVARIANT-SC Paper 1 Distinction
Schema Matching	Maps corresponding attributes using name similarity, structural embeddings, or instance samples.	Tests a complete candidate interpretation against coupled physical flow dynamics over time.
Domain-Constraint Matching	Uses declared static schema constraints or hand-written domain rules to prune mappings.	Learns latent process semantics directly from observed flow consistency and evaluates under shift.
Semantic Typing	Assigns known ontologies or labels to database columns, typically in isolation.	Recovers an interconnected process interpretation, measurement units, signs, and flow topology.
Relational Deep Learning	Learns representations on relational databases whose foreign keys and tables are given [2].	Solves the upstream prerequisite: recovering trustworthy common keys and process meaning.
Conservation-Aware ML	Embeds known differential equations into neural networks to improve forward prediction [3].	Inverts physical regularity to infer the semantic interpretation of the historical records themselves.
Traceability Standards	GS1 EPCIS 2.0 specifies how interoperable supply chain events should be structured [5, 6].	Reconciles messy, fragmented legacy evidence toward an auditable, canonical standard view.

A Defensible Novelty Claim

RIGOROUS QUALIFICATION

☆ CURRENT POSITION AFTER PRIOR-ART SCREENING

The differentiated contribution is the joint task of *physical-invariant-guided semantic recovery*, *global process consistency*, *evidence lineage*, and *explicit non-uniqueness* for heterogeneous supply-chain records. The cited fields contain important individual pieces, but not this complete learning target and evaluation object.

⊙ TOP-TIER PUBLICATION GATE

This is a concrete research hypothesis, not an unsupported "world-first" promise. The novelty becomes publication-grade only if the paper delivers the new task, a method that beats strong alternatives under unseen schemas, a rigorous account of ambiguity, and a reproducible benchmark. **Removing any two of these would make the work look like an integration system rather than a top-tier ML contribution.**

04 Why This Is Not Another Dashboard

SCIENTIFIC OBJECT VS. APPLICATION WRAPPER

THE FUNDAMENTAL DISTINCTION

A dashboard begins *after* meaning has already been supplied. INVARIANT-SC attacks the hidden prerequisite: it learns and audits the representation on which any trustworthy dashboard, optimizer, or carbon analysis must depend.

Dimension	Typical Dashboard / Commercial Platform	INVARIANT-SC (Paper 1 Scientific Object)
Input Data	Curated, pre-mapped, business-ready tables.	Heterogeneous raw records with uncertain semantics & links.
Core Operation	Aggregate, monitor, visualize, alert.	Infer, reconcile, and mathematically test shared process meaning.
System Output	Visual metrics, charts, scenarios, recommendations.	Canonical flow interpretation + lineage & unresolved alternatives.
Failure Mode	Manual developer exceptions or silent error propagation.	Abstain or expose competing interpretations under weak evidence.
Evaluation	UI usability and subjective business KPI improvements.	Recovery accuracy, cross-schema generalization, calibration.
Reusable Insight	A single company's deployed application configuration.	A general learning problem, method, identifiability theory & benchmark.

4.1 Proposed Paper Contributions

1. **A Benchmarkable Task:** Formalizes semantic and relational recovery under severe schema, unit, and organizational shift across disjoint enterprise ledgers.
2. **Learned-and-Verified Framework:** Couples local relational evidence with global physical-flow consistency (mass balance, temporal causality).
3. **Principled Characterization of Ambiguity:** Identifies mathematically when observations yield a unique interpretation versus an irreducible equivalence set.
4. **Reproducible Multi-Echelon Benchmark:** Standardized suite featuring controlled schema variation, missingness, reporting delays, and corruption.
5. **Downstream Error Propagation Evidence:** Quantifies how semantic recovery errors measurably compound into Scope 3 carbon, waste, and resilience metrics.

PRODUCT FEATURES ARE DELIBERATELY EXCLUDED

No procurement chatbot, KPI portal, universal carbon calculator, or multi-agent control room is needed for the first paper. A minimal visualization may illustrate results, but it is not the contribution.

05 Validation Plan & Real-World Value

EMPIRICAL RIGOR & INDUSTRIAL GROUNDING

5.1 What Would Make the Claim Credible

The evaluation must test scientific recovery rather than presentation aesthetics. Training and test splits must differ in database schemas, naming conventions, process topologies, organizations, and sensor noise. Baselines include modern LLM schema matchers [1], semantic typing engines, join discovery methods, relational deep learning [2], constraint solvers, and domain heuristics.

Empirical Test Dimension	Core Scientific Question Answered
Semantic & Relational Recovery	Did the system reconstruct the ground-truth roles, entity links, and process meanings across tables?
Unseen-Schema Generalization	Does physical flow guidance transfer robustly when database column labels and layouts shift completely?
Missing / Noisy / Delayed Records	Does the framework remain operational without treating every real-world operational delay as impossible?
Ambiguity & Calibrated Abstention	Does confidence drop predictably when available records permit multiple physically valid interpretations?
Downstream Error Propagation	Do improved recovered meanings measurably reduce error in carbon, circularity, and resilience metrics?

5.2 Real Issues Solved

- **Scope 3 Lineage:** Creates auditable activity data trails prior to applying emission factors [7].
- **Cross-Tier Contradictions:** Uncovers clashes across procurement, shop-floor, warehouse, and return logs.
- **Zero-Remapping Overhead:** Eliminates continuous manual re-engineering upon supplier software upgrades.
- **Stock-and-Flow Substrate:** Provides a unified physical baseline for shortage and disruption studies.
- **Defensible Claims:** Pinpoints exactly where enterprise evidence is mathematically insufficient.

5.3 Honest Scientific Limits

- **Unobserved Data:** Physical laws cannot recover unobserved data; some interpretations remain equivalent.
- **External Factors:** Emission factors and accounting rules remain strictly external to the model.
- **Social Dimensions:** Does not infer human labor conditions or supplier ethics from physical flows.
- **Real-World Transfer:** Synthetic benchmarks establish control, but real-world enterprise validation is vital.

STRATEGIC VENUE ALIGNMENT

ICLR fit is strongest if the paper leads with *representation learning*, *structured induction*, *cross-schema generalization*, and *identifiability theory*. If the theory or generalization claim does not survive, data-management venues (VLDB, SIGMOD) represent a natural home rather than overstretching the work into a dashboard paper.

06 Decision Summary & References

EXECUTION ROADMAP & CITATIONS

FACULTY BOARD RECOMMENDATION

Proceed with Paper 1 as **INVARIANT-SC**. Maintain exactly one headline scientific contribution: *auditable semantic recovery from physical flow consistency*. Anchor carbon, circularity, inventory, and resilience as coordinated downstream evaluations of the identical recovered model. Keep evidence and ambiguity subordinate to the core.

6.1 What Success Looks Like at Review Time

1 New Machine Learning Task
Reviewers recognize a fundamental structured learning problem, not a repackaged sustainability app.

2 Generalization Under Shift
Transfers robustly to unseen enterprise conventions because supervision is tied to immutable physics.

3 Calibrated Ambiguity
Exposes degenerate or non-unique interpretations openly rather than masking uncertainty with scores.

4 Unified Substrate
A single flow representation powers multiple downstream sustainability analyses without siloed ETL.

6.2 Selected Public References

[1] **N. Seedat and M. van der Schaar.** Bootstrapping Self-Improvement of Language Model Programs for Zero-Shot Schema Matching. ICML 2025 (PMLR).

[2] **M. Fey et al.** Position: Relational Deep Learning - Graph Representation Learning on Relational Databases. ICML 2024 (PMLR).

[3] **D. Hansen et al.** Learning Physical Models that Can Respect Conservation Laws. ICML 2023 (PMLR).

[4] **S. Chang et al.** Learning Production Functions for Supply Chains with Graph Neural Networks. AAAI 2025.

[5] **NIST.** Enabling and Using Traceability and Data Linking for Sustainable and Efficient Supply Chains. NIST Project Guidance.

[6] **GS1.** Electronic Product Code Information Services (EPCIS) Standard, version 2.0.1. Official Standard.

[7] **GHG Protocol.** Corporate Value Chain (Scope 3) Accounting and Reporting Standard. Official Guidance.

CONFIDENTIAL DISCLOSURE NOTE

This faculty brief intentionally omits internal algebraic formulas, tensor program representations, exact loss formulations, proof strategies, synthetic benchmark-generation recipes, acceptance thresholds, and private empirical results.

Novelty statement current to 2 September 2026: The proposal is strongly differentiated at the level of the joint learning task described here; final novelty and venue claims remain contingent on a systematic review, successful implementation, and comparative empirical evidence.